

Parvez Akhtar

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PROFESSIONAL SUMMARY

Senior GenAI Engineer with 5+ years shipping production LLM systems and agentic AI. Builds multi-step AI agents with multi-provider LLMs (OpenAI, Anthropic), advanced prompt engineering, agent context management, and eval-driven iteration (RAGAS, CI/CD gates). Reduced inference costs ~90% for production extraction and billing engines. Works daily across Python and TypeScript.

TECHNICAL SKILLS

GenAI & Agents: AI Agents & Multi-Agent Orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK, LangChain), Agent Context Management (sub-agents, retrieval-based RAG, sliding context windows), Multi-Step Workflow Orchestration, MCP (Model Context Protocol), Tool/Function Calling, Structured Outputs, Agent Memory & Human-in-the-Loop, Guardrails & Prompt-Injection Defense, Prompt Engineering & Programmatic Optimization (DSPy), RAG (Retrieval-Augmented Generation; Hybrid Search, Re-ranking), LlamaIndex, Fine-tuning (LoRA/QLoRA, Hugging Face), Embeddings, Vector DBs (Qdrant/PGVector).

Languages: TypeScript, Python, SQL, Java.

Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Spring Boot, FastAPI, Django.

Cloud & Infra: AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, Kubernetes, Docker, Terraform, Temporal.io.

Data Engineering: Snowflake Cortex, Apache Airflow, Kafka, Elasticsearch, MongoDB, Redis, PostgreSQL.

LLMOps & Evaluation: LLM Tracing, Evaluation & Benchmarking (Langfuse, OpenTelemetry, RAGAS), CI/CD Eval Gates, SigNoz, ELK Stack, CloudWatch, Prometheus.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Remote

Senior AI/ML Engineer, Lead Architect

04/2025 – Present

- As **Lead Architect**, drove cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous business problems into production architectures.
- **Agentic Automation:** Orchestrated a serverless **multi-agent** framework for Salesforce automation that coordinates **multi-step workflows** via **tool/function calling** over **MCP (Model Context Protocol)** and **swarm-based orchestration**. Cut idle compute via a pure **pay-per-execution** model (saving ~\$45k/year).
- **Client: VTS (current), Lease Abstraction.** Built an agentic **document-intelligence** pipeline that abstracts structured terms from commercial real-estate leases with **structured-output validation**. **Analyzed failure modes** behind a **10-point accuracy regression** and redesigned prompt strategy and image preprocessing to recover the full drop, restoring field-level extraction to its ~**88% baseline** and unblocking downstream automation.
- **Client: Outcomelabs.ai (GenAI Platform).** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **"Strategy Router"** that classifies each field by complexity and routes ~60% to deterministic regex/rule-based parsers, reserving the LLM for the rest, cutting token spend ~40% with no measurable accuracy loss.
- **Search & Ranking (applied research):** Implemented a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5** by ~**15 points (0.71 → 0.86)**.
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing fine-tuning strategies for cost efficiency.
- **Core Engineering:** Constructed an entity matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

Kuttl (Incubated at IIM Lucknow)

Remote

Founding Engineer

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the entire codebase and infrastructure, ensuring a **zero-downtime handover** of the platform to the incoming engineering team.
- Built the core backend and AI infrastructure from scratch, scaling the platform to handle peak loads while maintaining strict startup cost constraints.
- **Fine-tuned Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction tasks, achieving **98% schema compliance** while reducing per-token inference costs by **90%** and latency by 300ms.
- Engineered a distributed document parsing pipeline using **FastAPI** and **Temporal.io**, processing 50k+ daily messages. Enforced resilient retry logic and dead-letter queues to ensure no data loss.

- Operationalized the ML lifecycle with **automated LLM evaluation benchmarks** via **CI/CD eval gates (Ragas)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments that fell below threshold and catching quality regressions before production.
- Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** during high-concurrency bursts.

Amplicomm

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.